By the end of the exercises, you will be able to:

- Specifically extract digits from numbers using floor division and modulus to sum or reverse digits.
- Measure and calculate areas and distances applying appropriate formulas.
- Accept user input and convert temperature units accurately.
- Report results clearly using formatted output statements.
- Troubleshoot simple arithmetic operations in Python independently.

1) You are developing a simple reception desk system for a community centre. When a visitor arrives, the system needs to log their info and age to offer them a personalised greeting and direct them to suitable activities.

🪄 Task: Write a Python program that asks the visitor for their name and age, then prints a friendly greeting such as:

"Hello Sarah, welcome to the Community Centre! At 25, you're eligible for our young adults programme."

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))
print(f"Hello {name}, welcome to the Community Centre! At {age}, you're eligible for our young adults programme.")
```

```
Enter your name: Sarah
Enter your age: 25
Hello Sarah, welcome to the Community Centre! At 25, you're eligible for our young adults programme.
```

```
name=input("Enter your name:")
age=input("Enter your age:")
print(f"Hello {name},welcome to the Community Centre! At {age}, you're eligible for our young adults programme.")
```

```
Enter your name:Sarah
Enter your age:25
Hello Sarah,welcome to the Community Centre! At 25, you're eligible for our young adults programme.
```

```
Start coding or generate with AI.
```

2) You work for a home renovation company that installs floor tiles. A customer wants to tile their kitchen floor and needs to know how many square meters of tiles to buy.

🪄 Task: Write a Python program that asks the customer for the length and width of their kitchen (in meters), then calculates and displays the total area that needs to be tiled.

💬 Example Output: "To tile a kitchen of 3.5m x 2.8m, you will need 9.8 square meters of tiles."

🎯 Bonus Challenge: Each tile costs £12 per square meter. Add a cost calculator. Format the final message to include both the total area and the total cost:

"You will need 9.8 square meters of tiles. The total cost will be £117.60.

```
length = float(input("Enter the length of kitchen: "))
width = float(input("Enter the width of kitchen: "))

area = length * width
cost = area * 12

print(f"To tile a kitchen of {length}m x {width}m, you will need {area:.1f} square meters of tiles.")
print(f"You will need {area:.1f} square meters of tiles. The total cost will be £{cost:.2f}.")
```

```
Enter the length of kitchen: 3.5
Enter the width of kitchen: 2.8
To tile a kitchen of 3.5m x 2.8m, you will need 9.8 square meters of tiles.
You will need 9.8 square meters of tiles. The total cost will be £117.60.
```

3) You're helping build a basic weather app for travellers. One feature allows users to see today's temperature in either Celsius or Fahrenheit, depending on their preference.

🪄 Task: Write a Python program that asks the user for today's temperature in Celsius, then converts it to Fahrenheit using the formula: Then print the converted temperature in a user-friendly sentence.

$$°F = \frac{9}{5} \times °C + 32$$

```
# Ask user for temerature in Celsius
Celsius=float(input("Enter today's temerature in Celsius"))
# Convert using formula: F=(9/5*C)+32
# Print result in a friendly sentence
print(f{Celsius} in Celsius is {(9/5*Celsius)+32} in Fahrenheit)
```

```
Enter today's temperature in Celsius: 25
25.0 in Celsius is 77.0 in Fahrenheit
```

```
Start coding or generate with AI.
```

4) You are helping design a basic travel guide that helps users convert distances between miles and kilometres. For now, your app will do one conversion at a time, depending on which version of the program they run.

✨ Version A: Convert Miles to Kilometres 🪄 Task: Write a Python program that: Asks the user to enter a distance in miles Converts it to kilometres using the formula: **km=miles×1.60934**

Prints the converted distance.

✨ Version B: Convert Kilometres to Miles 🪄 Task: Write a Python program that: Asks the user to enter a distance in kilometres Converts it to miles using the formula: **miles=km÷1.60934**

Prints the converted distance.

```
# Ask user to  enter a distance in miles
miles=float(input("Enter the distance in miles"))
# Convert the miles into kilometers using formula: km=miles*1.60934
Km=miles*1.60934
# Print the converted distance.
print(f{miles}) miles is {Km} in Kilometers)
```

```
Enter the distance in miles: 13.1
13.1 miles is 21.082354 in Kilometers
```

```
Start coding or generate with AI.
```

5) You're helping a theme park design a fun feature for visitors. When they enter a 3-digit number, your program will add the digits together to tell them their "lucky number."

🪄 Task: Write a Python program that: Asks the user to enter a 3-digit number

Separates the hundreds, tens, and units digits using floor division (//) and modulus (%)

Adds the digits together

Prints the result as their "lucky number"

```
enter three digit number: 258
15
```

```
# Ask visitor to input a 3-digit number
number=int(input("Enter a 3-digit number"))
# Separate digits using floor division and modulus
hundreds=number//100
tens=(number//10)%10
units=number%10
# Add the digits together
lucky_number=hundreds+tens+units
# Print result
print(f"Your lucky number is {lucky_number}")
```

```
Enter a 3-digit number258
Your lucky number is 15
```

```
Start coding or generate with AI.
```

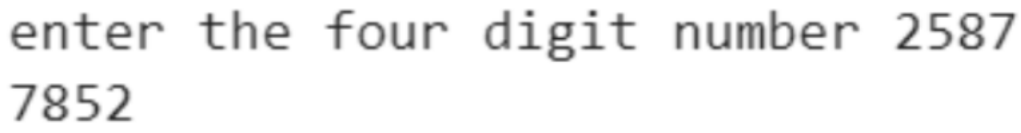
6) You're helping a museum design an escape room game. When players find a 4-digit number, they must reverse it using your program to unlock the next clue.

🧪 Task: Write a Python program that:

Asks the user to enter a 4-digit number

Uses floor division (//) and modulus (%) to separate the digits

Reverses the digits and prints the result as the "secret code




```
# Ask player to input a 4-digit number
number=int(input("Enter a 4-digit number:"))
# Separate the digits by using floor division and modulus
digit1=number//1000
digit2=(number//100)%10
digit3=(number//10)%10
digit4=number%10
# Reverse the digits
reversed_number=digit4*1000+digit3*100+digit2*10+digit1
# Print the reversed number
print(f"The secret code is {reversed_number}")
```

```
Enter a 4-digit number:2587
The secret code is 7852
```

```
Start coding or generate with AI.
```